Worksheet — Spring Force Simulation

Use with *Simulation - Spring force.html* · goes with Station 3 and the Forces investigation (spring)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/05-spring-force** in a browser (offline: *Simulation - Spring force.html*). **Hang masses**: add 50 g at a time and read the pointer on the ruler. **Push / pull**: stretch or squash a spring on a table. The **Tasks** tab has the same four parts as this sheet; write your **explanations** here.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 10 min) · **Part 3** homework (≈ 30 min) · **Part 4** checkpoint: hysteresis and the elastic limit (≈ 15 min).

Use g = 9.81 N kg^{-1}. Extension x = pointer reading − unloaded reading. Always convert x to metres for k.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. A mass hangs at rest on the spring. Is the spring force bigger than, equal to or smaller than its weight?			
2. You double the load from 100 g to 200 g. What happens to the extension?			
3. You load the soft spring (limit ≈ 3 N) with 400 g, then unload it. What happens?			
4. You squash the spring by 6.0 cm. Which way does the spring push the block?			

5. Draw the free-body diagram of a 200 g mass hanging at rest on the spring.

Free-body diagram

6. The investigation spring has k = 25.0 N m^{-1}. Calculate the extension (in cm) with 200 g hanging.

Working

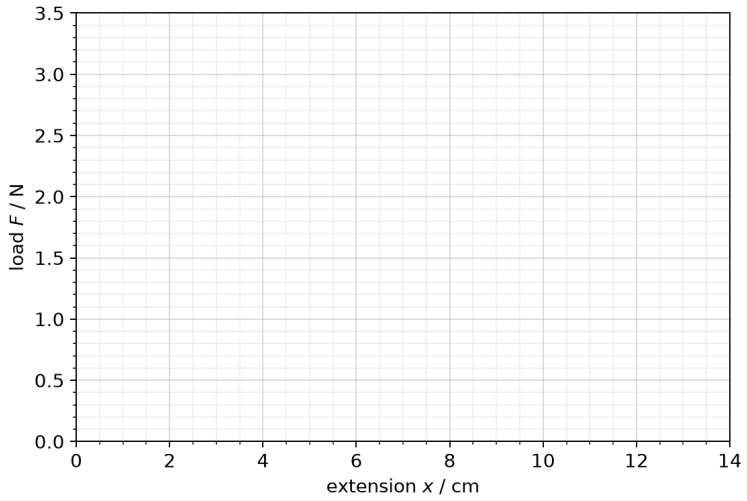
x = _____ cm Sim check: _____ cm

Part 2 — During the lab: Station 3 / Forces investigation

Investigation spring, reading uncertainty ON. Record the sim values and your own measurements.

mass m / g (±1)	load $F = mg$ / N	sim pointer / cm	sim x / cm (±0.1)	our x / cm (±0.1)
100				
150				
200				
250				
300				

7. Plot F against x with error bars (±0.1 cm). Draw the best-fit line and the steepest and shallowest lines through the error bars of the first and last points.



8. k = _____ ± _____ N m^{-1} ($\Delta k = \frac{1}{2}(k_{max} - k_{min})$)

9. Mystery bag: x = _____ cm ⇒ $F = kx$ = _____ N ⇒ m = _____ kg. Balance: _____ kg

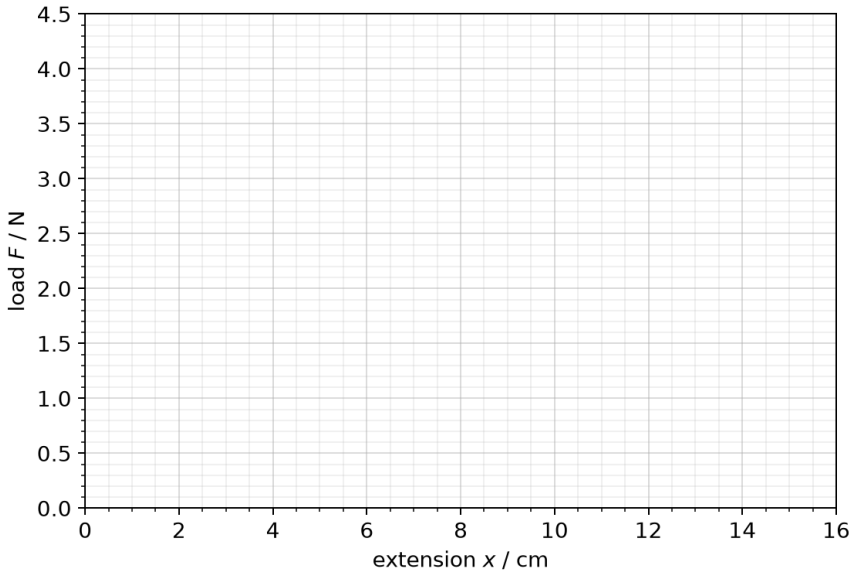
10. Would your method still work for a 5 kg load? Explain.

Part 3 — Homework: analyse

Station 3 spring, uncertainty ON. Record 100 g to 400 g in 50 g steps.

m / g	100	150	200	250	300	350	400
F / N							
x / cm							

11. Plot *F* against *x* with error bars and find $k \pm \Delta k$.



k = _____ ± _____ N m^{-1}

12. The investigation spring is stretched by 6.0 cm on a table. Calculate the spring force and give its direction. What does the minus sign in $F_H = -kx$ mean?

Working

13. The stiff spring (k = 40.0 N m^{-1}) stretches 4.5 cm when an object hangs from it. Calculate the mass of the object.

Working

14. Sketch the *F*–*x* graph of a spring loaded past its limit of proportionality. Mark the limit of proportionality. Explain why an overloaded spring is no longer a good force meter.

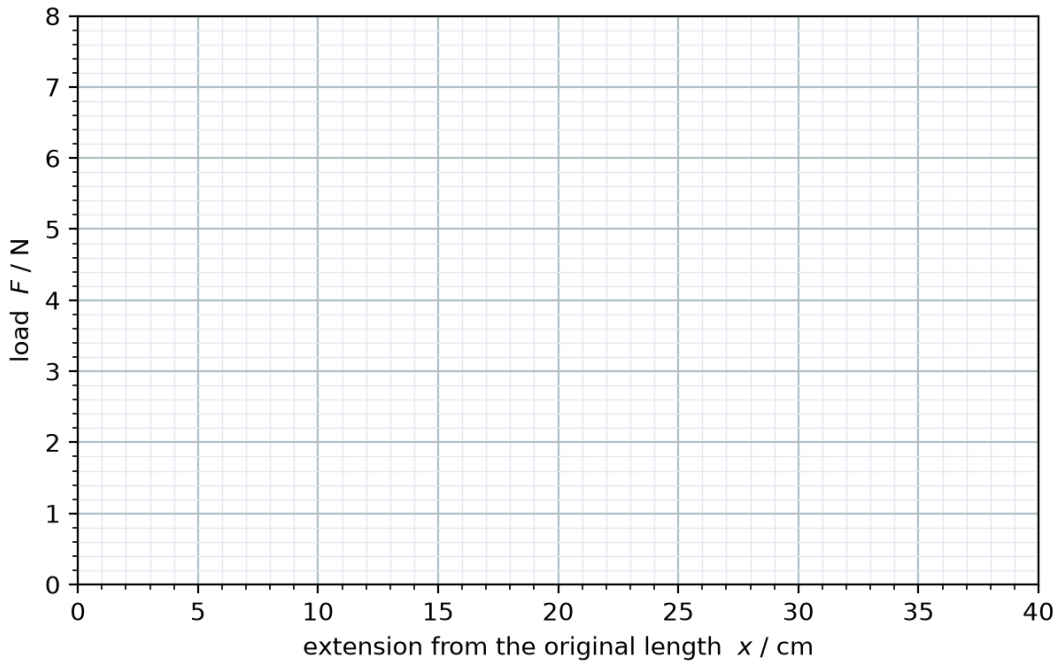
Sketch and explanation

Part 4 — Checkpoint: hysteresis and the elastic limit

Simulation: mode **Load & unload**. Spring: Investigation spring (k = 25.0 N m^{-1}, elastic limit **E** at 6.0 N). Load it in steps up to 800 g, then unload. Record the extension readout (pointer − 30.0 cm) x each time the load is at rest.

mass m / g	load $F = mg$ / N	loading: x / cm	unloading: x / cm
0			
200			
400			
600			
700			
800			(turning point)

15. Plot the loading points and the unloading points on the grid and join each set. Mark the elastic limit **E** and label the loading and unloading lines with arrows.



16. Read the **permanent extension** from your graph: _____ cm. Explain why the unloading line does not go back through the origin.

_____

_____

17. Spring: Rubber band. Load it to 500 g in 50 g steps, then unload to 0 g. Sketch the loop (no numbers needed). Use arrows for loading and unloading and **shade the area that equals the energy turned into heat**.

Sketch

18. From the readouts after the full cycle: energy stored = _____ J, energy given back = _____ J, so energy → heat = _____ J = _____ % of the energy stored.

19. This is called **hysteresis**. Explain why a rubber band that is stretched and released many times gets warm. Would you choose a material with a large or a small loop for the rubber mounts that stop a washing machine shaking? Why?